

# Front wing in ground effect beside a rotating wheel

Steady RANS setup and verification report

OpenFOAM v2606 (native) • `snappyHexMesh` + `simpleFoam` • case S1.42\_h0.13\_AOA8\_W0.63

**State of this report.** The setup, the mesh and the whole figure and reporting chain are complete and verified. **The solution is not.** This build stops at **88 SIMPLE iterations** of the 2000 configured, so every flow quantity in it — forces,  $C_p$ ,  $C_p T$ ,  $y^+$ ,  $Q$  — is a *partially converged* field and must not be quoted as a result. The residuals are still at  $10^{-2}$  (section 7). Mesh-side numbers (cell counts, layer coverage, `checkMesh`, timings) are final and can be relied on. To finish it: run the solve to completion and rebuild, see section 9.

## 1 Case at a glance

| Configuration   |                               | Mesh                         |            |
|-----------------|-------------------------------|------------------------------|------------|
| wing half-span  | $S/c = 1.42$                  | cells                        | 18 322 109 |
| ride height     | $h/c = 0.13$                  | background cells             | 123 624    |
| angle of attack | $8^\circ$                     | layer coverage               | 96.0 %     |
| wheel width     | $W/c = 0.63$                  | <code>checkMesh</code> clean | no         |
| Flow            |                               | Cost, 20 MPI ranks           |            |
| freestream      | 10.0 m/s                      | meshing                      | 20 min     |
| chord           | 0.075 m                       | solving                      | 12 min     |
| $Re_c$          | 46 793                        | per iteration                | 7.59 s     |
| $\rho$          | 1.1629 kg/m <sup>3</sup>      | iterations                   | 88         |
| $\nu$           | 1.60278e-05 m <sup>2</sup> /s | converged                    | no         |

**Headline result.** Force *coefficient-areas*, mean over the last 50 iterations.  $A_{\text{ref}} = 1 \text{ m}^2$  on purpose, so these are  $C_L \cdot A$  and  $C_D \cdot A$  in m<sup>2</sup> — the quantity the Diasinos papers tabulate. Negative lift is downforce.

|       | $C_L \cdot A$ [m <sup>2</sup> ] | $C_D \cdot A$ [m <sup>2</sup> ] | spread of $C_L \cdot A$ |
|-------|---------------------------------|---------------------------------|-------------------------|
| wing  | -0.007407                       | 0.000959                        | 22.64 %                 |
| wheel | 0.001337                        | 0.002344                        | 46.82 %                 |
| total | -0.006070                       | 0.003303                        | 28.23 %                 |

Mean  $y^+$ : wing 3.94, wheel 4.65, maximum 73.1.

## 2 Software and reproducibility

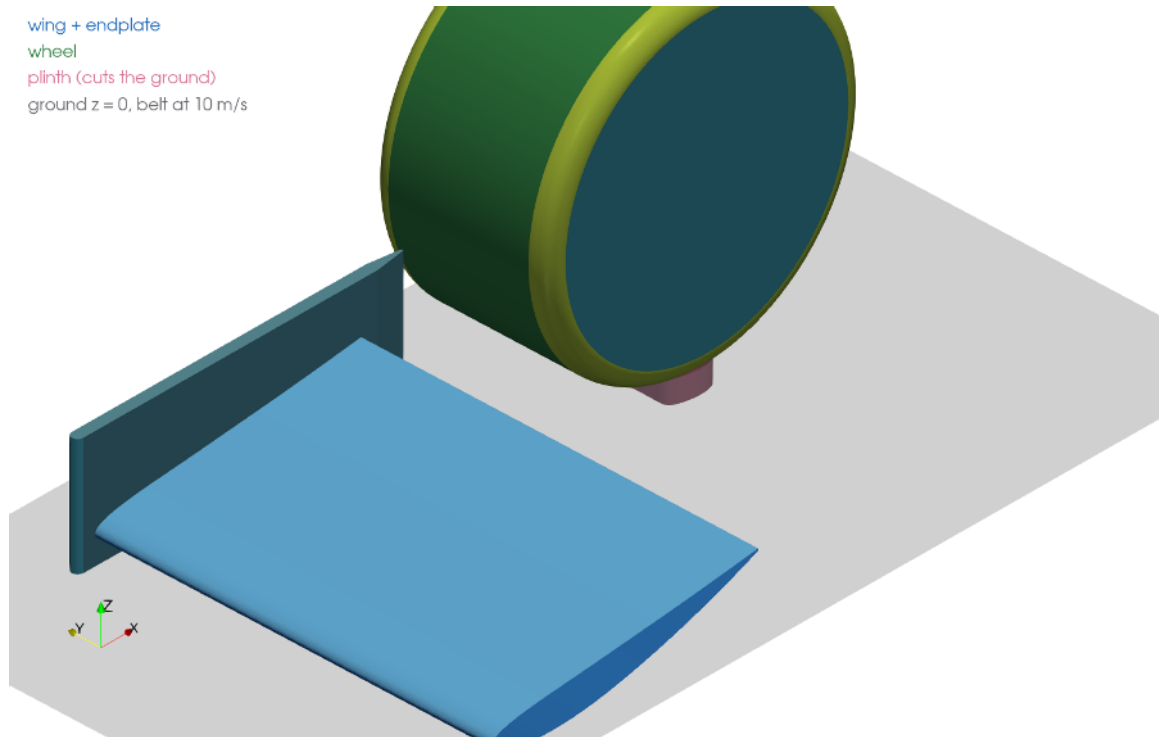
- **OpenFOAM v2606** (ESI/OpenCFD), built from source. Native only — no commercial solver, no in-house libraries.
- Mesh `snappyHexMesh`, solve `simpleFoam` (segregated, SIMPLEC), turbulence `kOmegaSST`.
- Geometry generated parametrically from the Diasinos specification; 420 configurations available.
- Everything else is `python3` standard library. No `numpy/pandas` needed to run a case.

To reproduce this case.

```
cp env/project.env.example env/project.env # set OPENFOAM_BASHRC and NP
cd Diasinos_Geometry_Generator && .venv/bin/python -m geometry_generator.generate
cd ../RANS_Simulations
python3 prepare_case.py --span 1.42 --height 0.13 --aoa 8 --width 0.63
cd S1.42_h0.13_AOA8_W0.63 && ./Allrun
```

### 3 Geometry

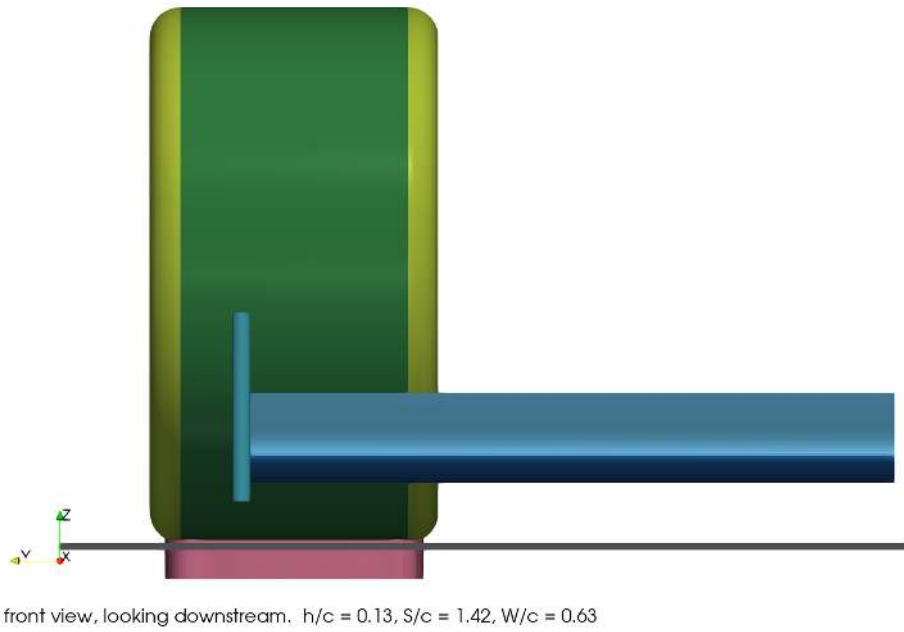
- Coordinate system:  $x$  streamwise (wheel centre at  $x = 0$ ),  $y$  spanwise outboard,  $z$  up with the ground at  $z = 0$ . Half model, symmetry at  $y = 0$ .
- Chord  $c = 75$  mm. Inverted NACA 4412, blunt trailing edge 0.5 mm. Wheel diameter  $1.17c$ , shoulder radius  $0.067c$ .
- **The generator writes millimetres**; everything in the simulation is metres. Allmesh scales by 0.001 on the way in.
- The *plinth* is a collar that deliberately pierces the ground plane (down to  $z = -6$  mm) so the contact patch is a clean cut instead of a tangential singularity.



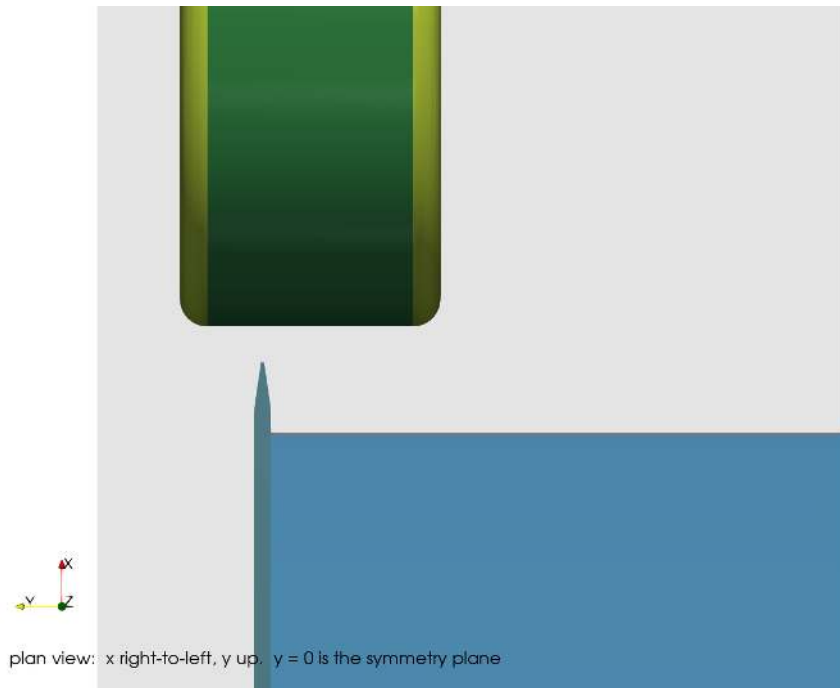
**Figure 1:** Geometry and the ground plane. The wing sits in ground effect ahead of and inboard of the wheel; the plinth passes through the ground.

#### Known geometry limitations.

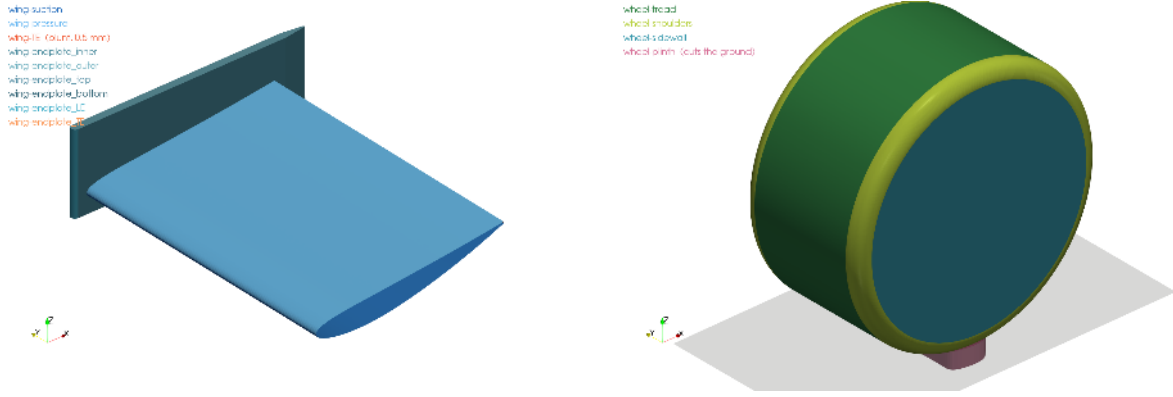
- The wing is **prismatic**, extruded between two spanwise stations. Geometrically exact for a constant section; it does mean the triangles have a very high aspect ratio, which **surfaceCheck** reports as poor quality. Not an error.
- **Two duplicate triangles** on the endplate outer edge — a defect in the generator. Two facets out of 1142; **snappyHexMesh** tolerates them. Reported by **Allmesh** and deliberately non-fatal.
- Both STLs consist of two disjoint parts (wing + endplate, wheel + plinth), so **surfaceCheck** says “not closed”. Expected: no **mode inside** refinement region is attached to either body.



**Figure 2:** Front view, looking downstream. The endplate (light blue) falls inside the wheel silhouette: at  $S/c = 1.42$  and  $W/c = 0.63$  the spanwise gap is  $-0.45c$ , i.e. wing and wheel overlap.



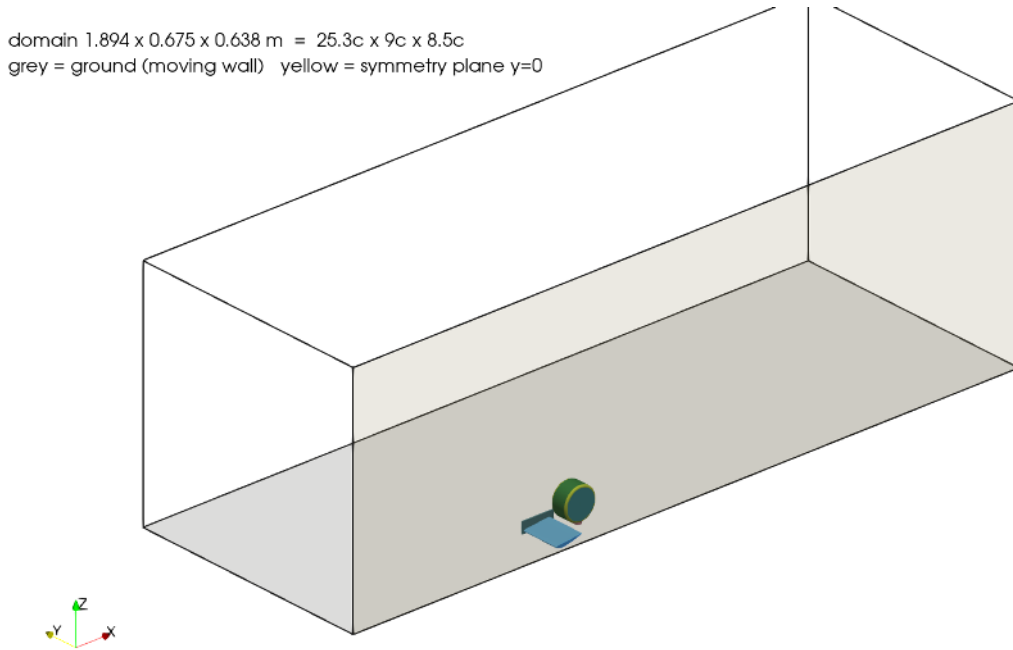
**Figure 3:** Plan view.  $y = 0$  is the symmetry plane.



**Figure 4:** Named STL regions. `snappyHexMeshDict` refers to these names directly to set refinement level and patch name, so renaming a region in the generator breaks the mesh setup.

- Wheel tessellation was raised from 180 to 900 circumferential segments, so the facet is 0.31 mm against the 0.293 mm cell. At 180 segments the facets were 1.53 mm — five times coarser than the cell, i.e. the mesh resolved the wheel as a 180-gon with a visibly wavy surface. *This did not improve the layer coverage on the tread* (63.4% before and after); it was done for geometric fidelity alone.

## 4 Domain and boundary conditions



**Figure 5:** Computational domain,  $1.894 \text{ m} \times 0.675 \text{ m} \times 0.638 \text{ m} = 25.3c \times 9c \times 8.5c$ . Grey: moving ground. Yellow: symmetry plane.

| patch / group | type     | $U$                         | $p$          | $k, \omega, \nu_t$ |
|---------------|----------|-----------------------------|--------------|--------------------|
| inlet         | patch    | fixedValue (1000)           | zeroGradient | turbulent inlet    |
| outlet        | patch    | pressureInletOutletVelocity | fixedValue 0 | inletOutlet        |
| symmetry      | symmetry | symmetry                    | symmetry     | symmetry           |
| side, sky     | patch    | slip                        | slip         | slip / calculated  |
| ground        | wall     | fixedValue (1000)           | zeroGradient | wall functions     |
| wingGroup     | wall     | noSlip                      | zeroGradient | wall functions     |
| wheelRotating | wall     | rotatingWallVelocity        | zeroGradient | wall functions     |
| wheelGround   | wall     | fixedValue (1000)           | zeroGradient | wall functions     |

- **Rolling road:** the belt translates downstream at the freestream speed, so the ground is at rest relative to the far-field air.
- **Rotating wheel:**  $\omega = -227.92$  rad/s about (0 1 0) through (0, 0, 0.043875 m). `rotatingWallVelocity` imposes  $\omega \times r$  without moving the mesh — correct for steady RANS. The sign follows from requiring the contact point to move with the belt:  $\omega_y = -U_\infty/R$ . Reversing it spins the wheel backwards and still converges happily.
- The **wheel-plinth** patch is in `wheelGround`, not `wheelRotating`: it is the collar through the ground and moves with the belt.
- **side** and **sky** are `patch`, not `wall`, on purpose — a wall there would enter the `meshWave` wall-distance field that `kOmegaSST` blends on, and would appear in the force groups.

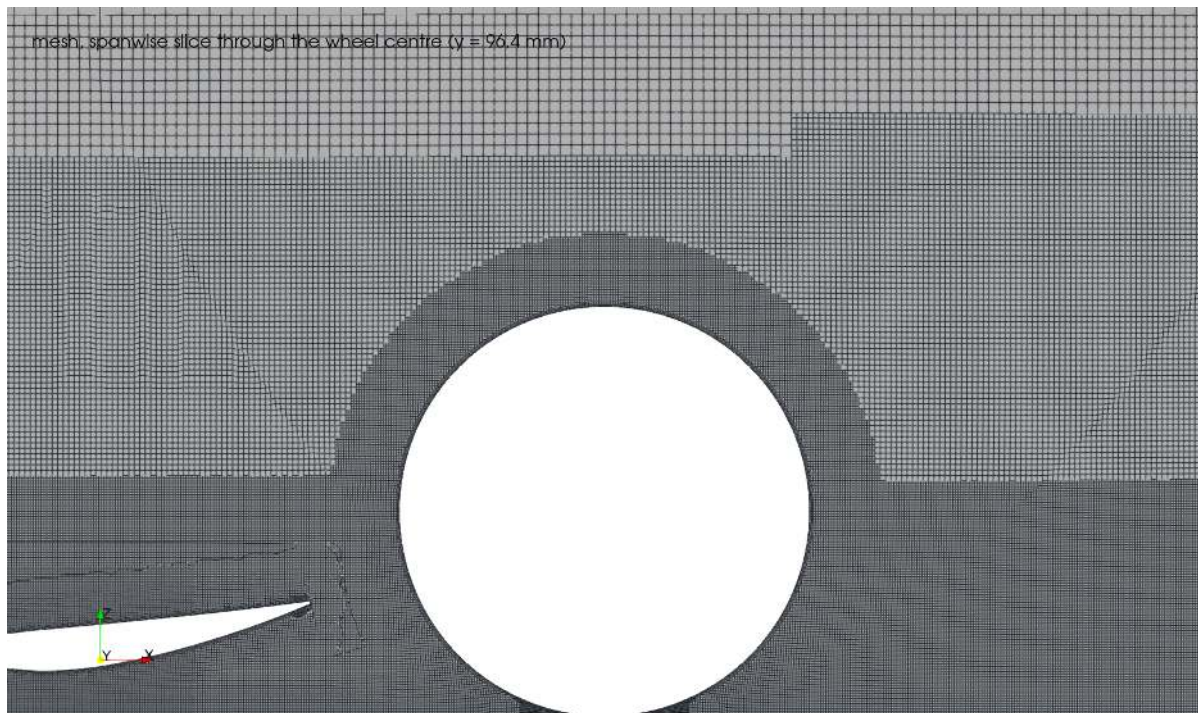
## 5 Mesh

- Background:  $101 \times 36 \times 34$  **cubic** cells of 18.75 mm ( $= 0.25c$ ), 123 624 cells. Cubic matters: each refinement level halves the edge, so the level-to-size table only holds from a cube.
- Level sizes: L4 1.17 mm, L5 0.586 mm, L6 0.293 mm, L7 0.146 mm.
- Per-region surface levels, distance refinement around both bodies (L5 within 15 mm, L4 within 30 mm), and nine `searchableBox` zones.
- Final mesh 18 322 109 cells.

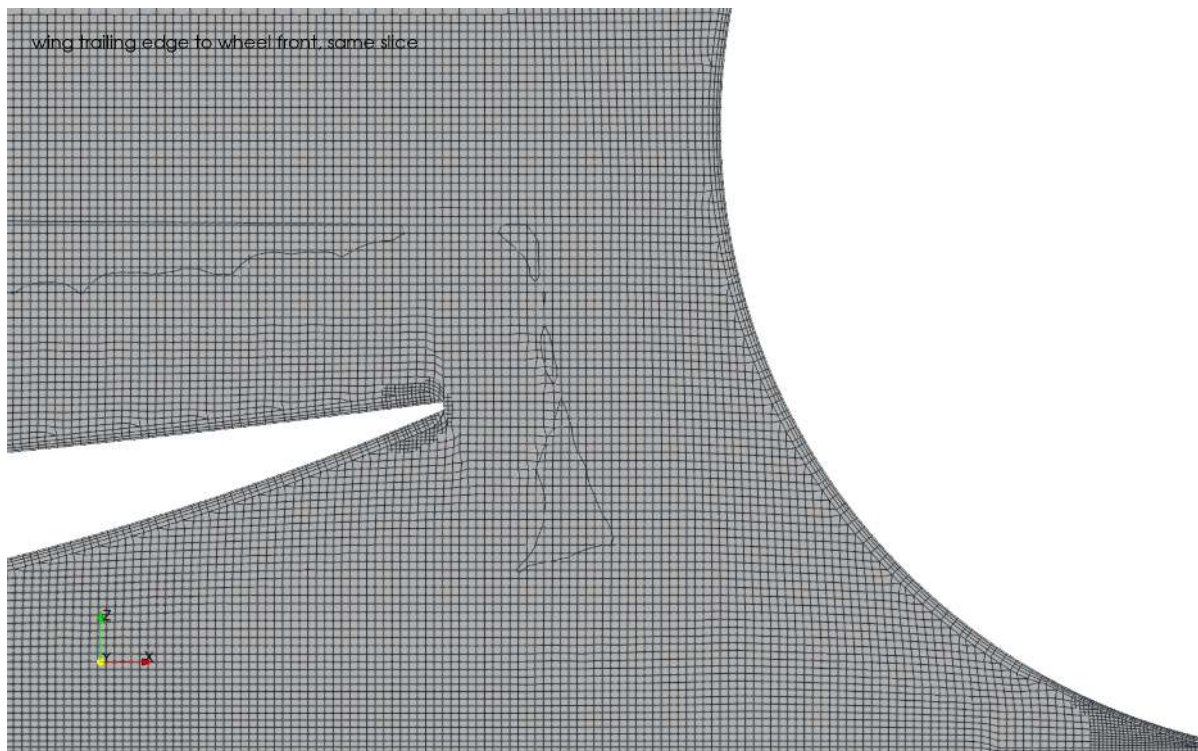
| refinement level | cells      | share [%] |
|------------------|------------|-----------|
| L0               | 110 876    | 0.6       |
| L1               | 19 040     | 0.1       |
| L2               | 519 518    | 2.8       |
| L3               | 592 509    | 3.2       |
| L4               | 4 302 360  | 23.5      |
| L5               | 11 579 533 | 63.2      |
| L6               | 1 194 917  | 6.5       |
| L7               | 3 356      | 0.0       |

### 5.1 Prism layers

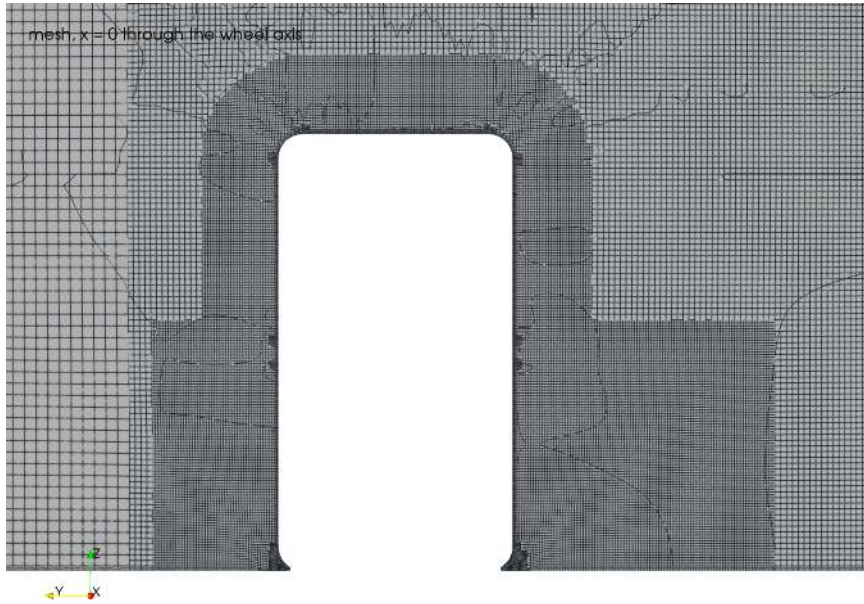
Four layers from 0.14 mm with expansion 1.3 (total 0.87 mm) on wing and wheel, three from 0.25 mm on the ground. This targets  $y^+ \approx 2.5$ , which is why the wall functions are the all- $y^+$  variants.



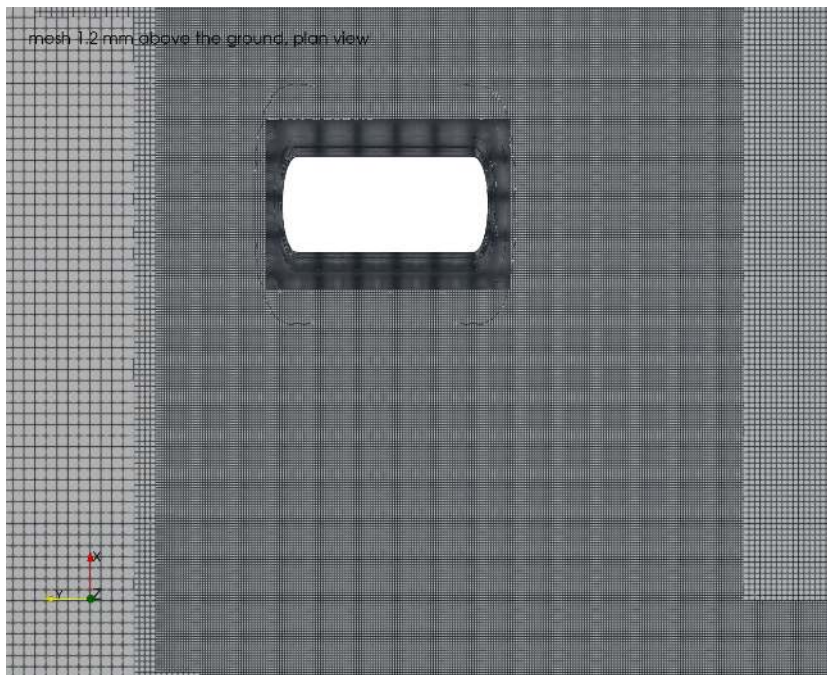
**Figure 6:** Mesh on a spanwise slice through the wheel centre. The nested refinement zones and the wake box are visible.



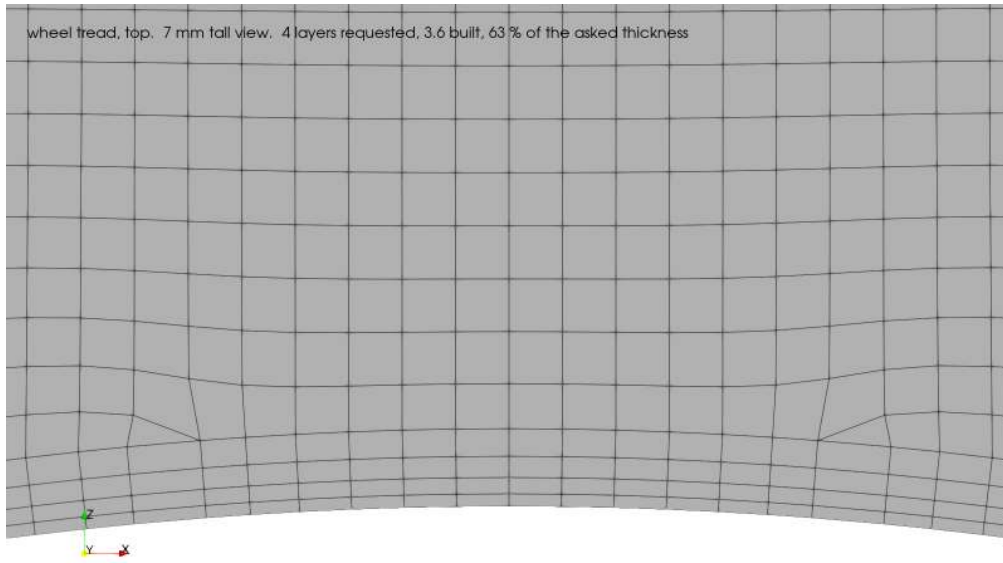
**Figure 7:** Same slice, wing trailing edge to wheel front — the region the whole study is about.



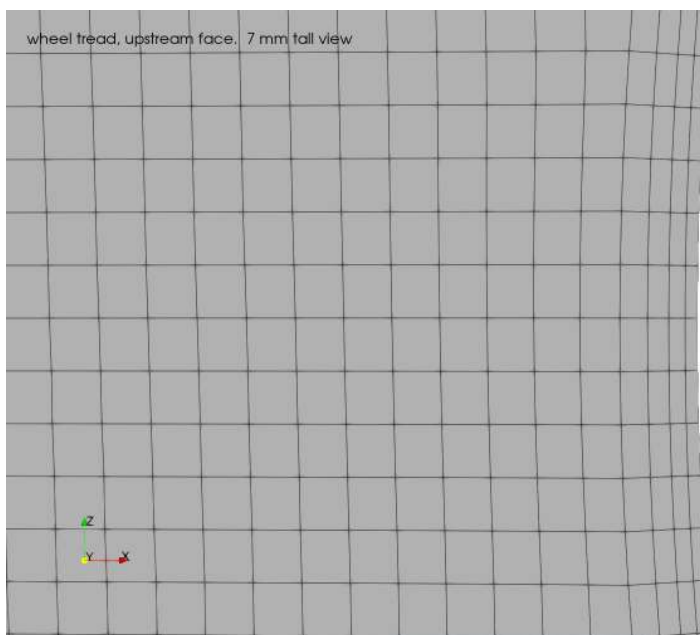
**Figure 8:** Slice through the wheel axis ( $x = 0$ ). Looking downstream, so  $y$  increases to the left.



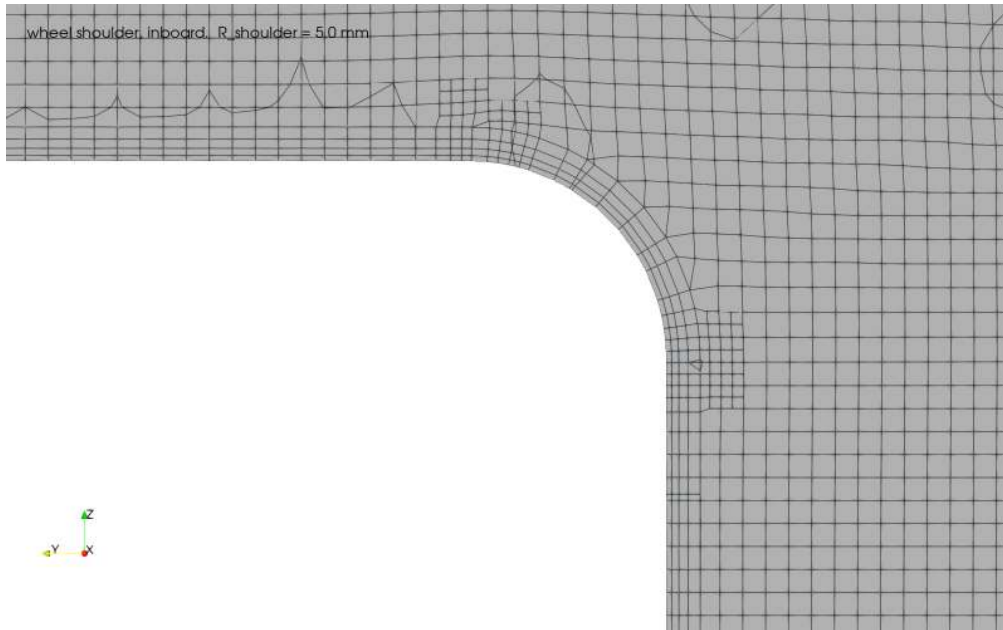
**Figure 9:** Mesh 1.2 mm above the belt. The contact-patch refinement box is the fine square under the wheel.



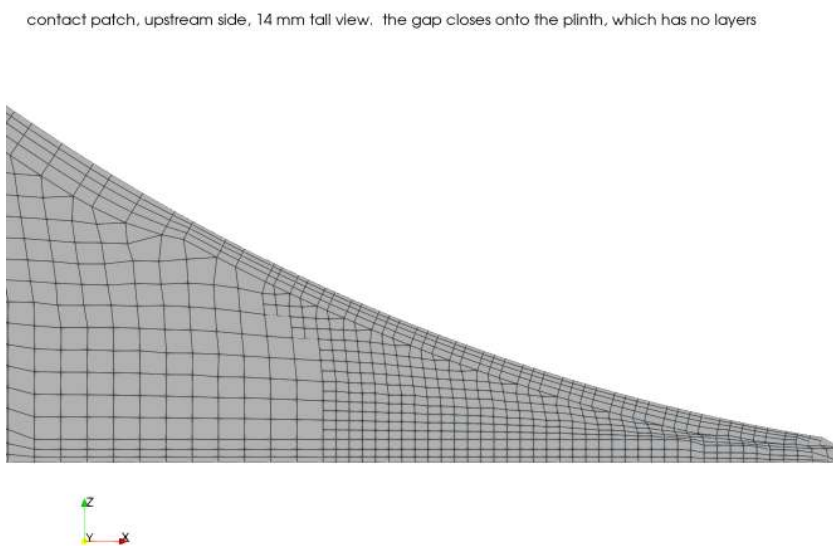
**Figure 10:** Wheel tread at the top. Prism layers on the 0.31 mm-tessellated tread.



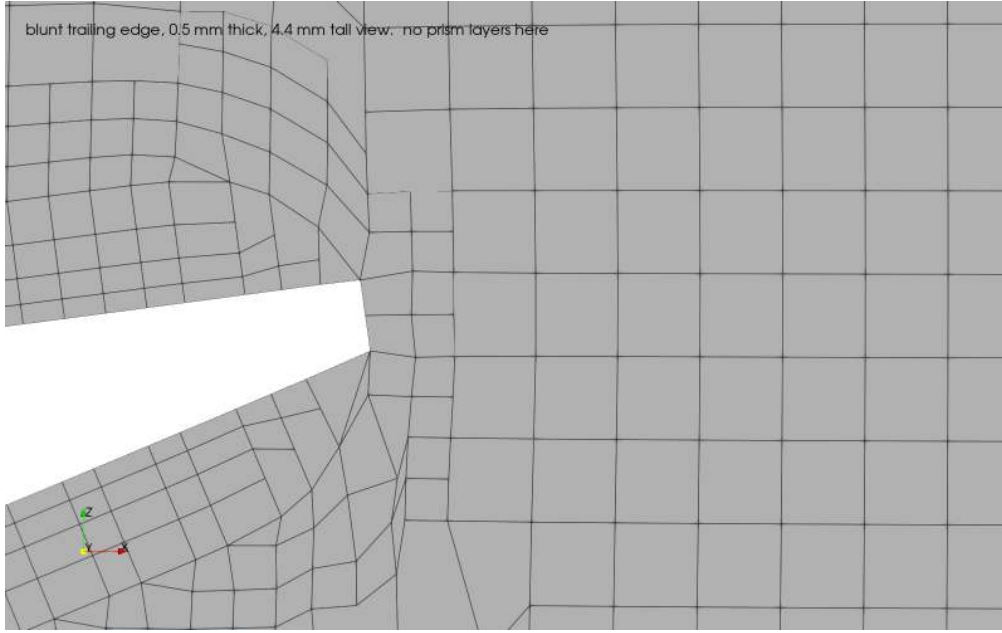
**Figure 11:** Wheel tread, upstream face.



**Figure 12:** Wheel shoulder, inboard. The 5 mm shoulder radius is the tightest convex curvature on the wheel.



**Figure 13:** Contact patch. The plinth cuts the ground plane at  $z = 0$ ; this is where the layers fight each other.



**Figure 14:** Blunt trailing edge, 0.5 mm thick. No prism layers — see below.

| patch                   | faces  | layers wanted | layers built | thickness [%] |
|-------------------------|--------|---------------|--------------|---------------|
| ground                  | 180851 | 3             | 2.92         | 92.3          |
| wing-suction            | 28951  | 4             | 3.88         | 95.1          |
| wing-pressure           | 27637  | 4             | 3.87         | 96.2          |
| <b>wing-TE</b>          | 746    | 2             | 0            | 0             |
| wing-endplate_inner     | 13234  | 4             | 3.45         | 79.1          |
| wing-endplate_outer     | 14498  | 4             | 3.71         | 94.3          |
| wing-endplate_top       | 3256   | 4             | 2.85         | 76            |
| wing-endplate_bottom    | 3256   | 4             | 2.85         | 76.4          |
| wing-endplate_LE        | 1356   | 4             | 3.87         | 97.3          |
| <b>wing-endplate_TE</b> | 212    | 2             | 0            | 0             |
| wheel-tread             | 38861  | 4             | 3.95         | 84.3          |
| wheel-shoulders         | 26056  | 4             | 3.88         | 89.9          |
| wheel-sidewall          | 33128  | 4             | 4            | 100           |
| <b>wheel-plinth</b>     | 902    | 2             | 0            | 0             |

### Reading the layer table.

- Overall coverage 96.0 %.
- The patches in **bold** got **no layers at all**. These are **wing-TE**, **wing-endplate\_TE** and **wheel-plinth** — sharp convex edges and the plinth/ground intersection. Asking for fewer and thinner layers does not change it: the limit is topology, not thickness.
- **This is accepted.** **snappyHexMesh** is known for incomplete layer coverage at sharp edges and where two layered surfaces meet. All three patches sit in L7 cells (0.146 mm), so the near-wall spacing is comparable to the layered regions anyway.
- The number to judge the near-wall treatment by is  $y^+$  (section 8), not the layer count.

## 5.2 Mesh quality

Quality limits are deliberately relaxed (**maxNonOrtho** 75, **maxInternalSkewness** 6, **minTetQuality**  $-10^{30}$ ), carried over from the previous HELYX setup. Enforcing a positive tet quality with absolute layer thicknesses on curved surfaces costs most of the layer coverage.

**checkMesh** clean: **no** (4 findings).

- Error in face tets: 248 faces with low quality or negative volume decomposition tets.

- Cells with small determinant ( $< 0.001$ ) found, number of cells: 531
- Concave cells (using face planes) found, number of cells: 154366
- Faces with small interpolation weight ( $< 0.05$ ) found, number of faces: 80

These are the expected consequence of the relaxed limits. The solver runs stably on this mesh; if it ever diverges, `meshQualityDict` is the first thing to question.

## 6 Physics and numerics

|                        |                                                  |
|------------------------|--------------------------------------------------|
| turbulence model       | kOmegaSST                                        |
| $\nu_t$ wall function  | nutUSpaldingWallFunction (continuous all $y^+$ ) |
| $k$ wall function      | kLowReWallFunction                               |
| $\omega$ wall function | omegaWallFunction                                |
| momentum convection    | bounded Gauss linearUpwindV grad(U)              |
| turbulence convection  | bounded Gauss upwind (first order, deliberate)   |
| gradients              | cellLimited Gauss linear 1                       |
| laplacian              | Gauss linear corrected                           |
| wall distance          | meshWave                                         |
| pressure-velocity      | SIMPLEC (consistent yes)                         |
| $p$ solver             | GAMG, tolerance $10^{-7}$ , relTol 0.01          |
| $U, k, \omega$         | smoothSolver GaussSeidel                         |
| relaxation             | $U$ 0.9, $k$ and $\omega$ 0.7                    |
| residualControl        | $p, U$ : $10^{-5}$ ; $k, \omega$ : $10^{-4}$     |
| initialisation         | potentialFoam, divergence-free start             |

**Turbulence convection is first order on purpose.** Second-order  $k/\omega$  transport is the usual reason residuals stall on a mesh of this kind, and the effect on the integrated forces is small. Momentum convection is second order.

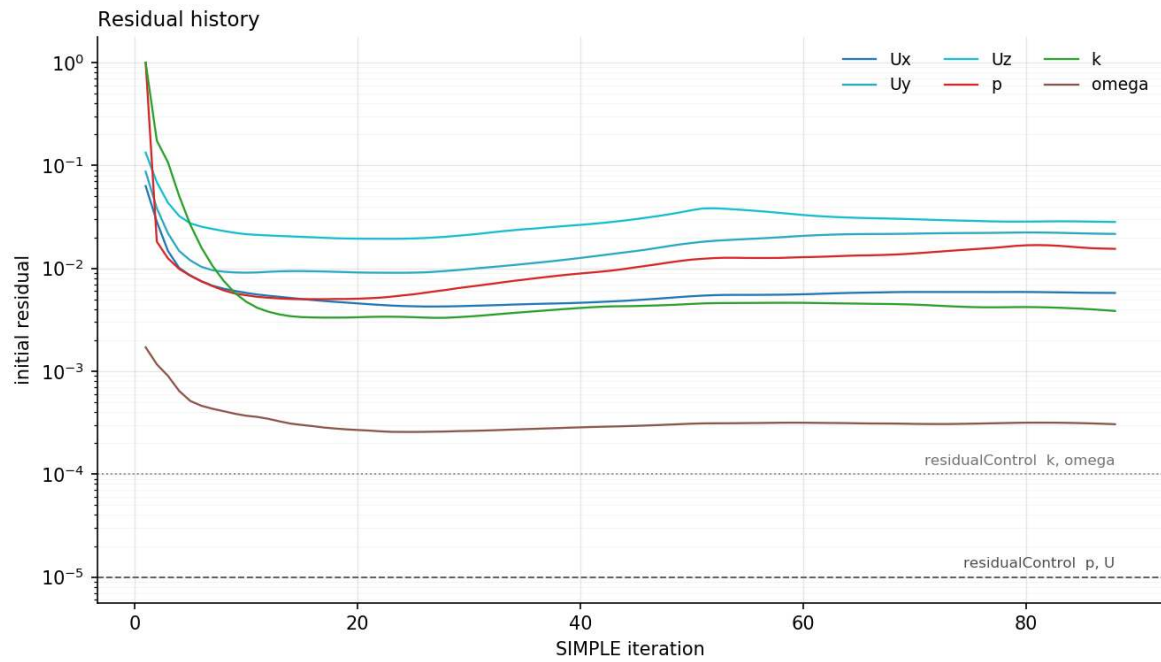
**Differences against the previous HELYX setup.** Relevant whenever the two are compared:

- **No curvature correction** (HELYX used Menter–Smirnov). The endplate and wheel vortex cores will diffuse somewhat faster.
- **No separation fix** (HELYX used the Rumsey form). Separation onset on the suction side may differ.
- The adaptive wall treatment *is* reproduced, by the all- $y^+$  wall functions above.
- HELYX ran a coupled solver pseudo-transiently; this is segregated and steady, so iteration counts are not comparable.

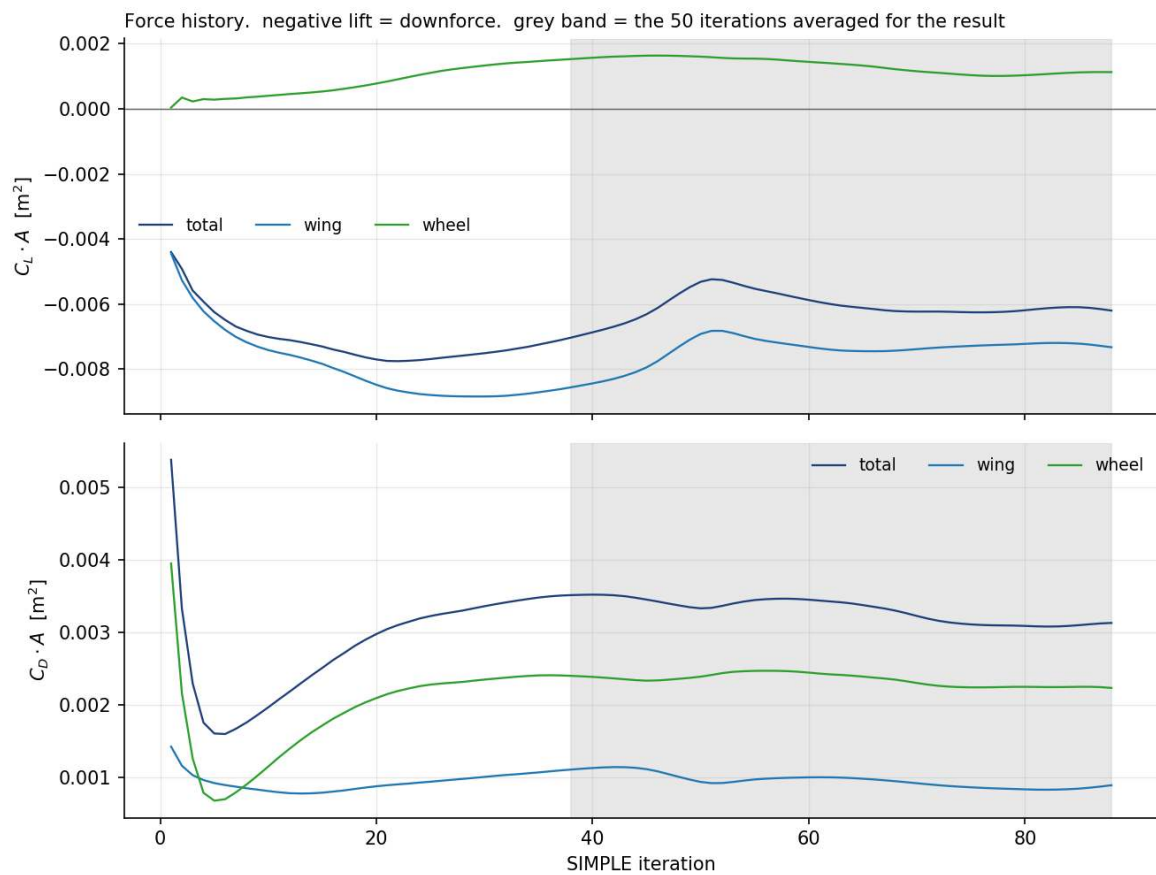
## 7 Convergence

**How to judge this.**

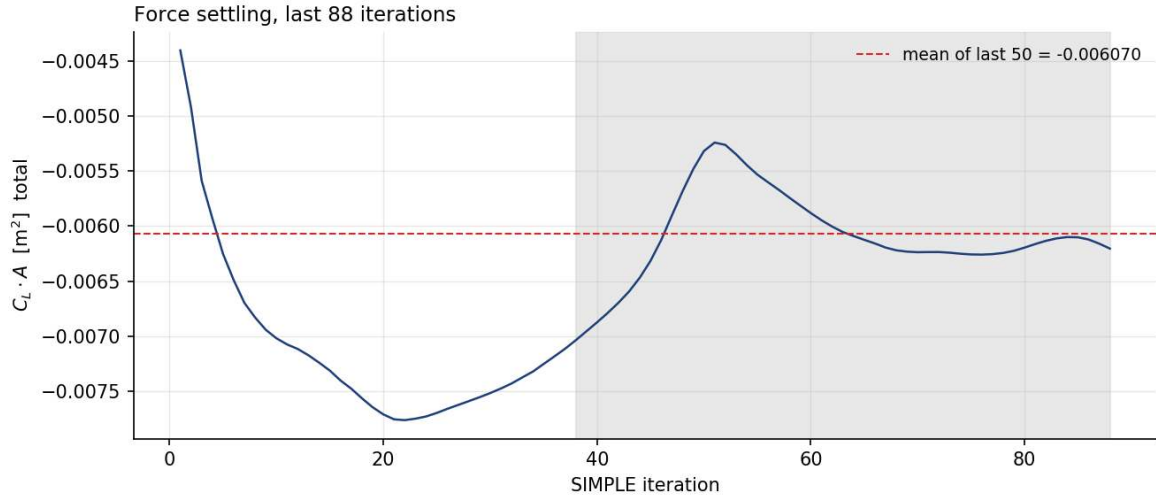
- **residualControl** satisfied: **no**.
- Final initial residuals:  $p$  1.55e-02,  $U_x$  5.77e-03,  $U_z$  2.82e-02,  $k$  3.85e-03,  $\omega$  3.05e-04.
- **The force spread matters more than the residual.** A converged residual with  $C_L \cdot A$  still swinging by several percent means an unsteady flow being forced into a steady solve. Here the total spread over the averaging window is 28.23%.
- Residual *fields* are written as well, so a stalled case can be opened in ParaView to see *where* it will not settle.



**Figure 15:** Initial residuals per SIMPLE iteration, with the `residualControl` thresholds marked.



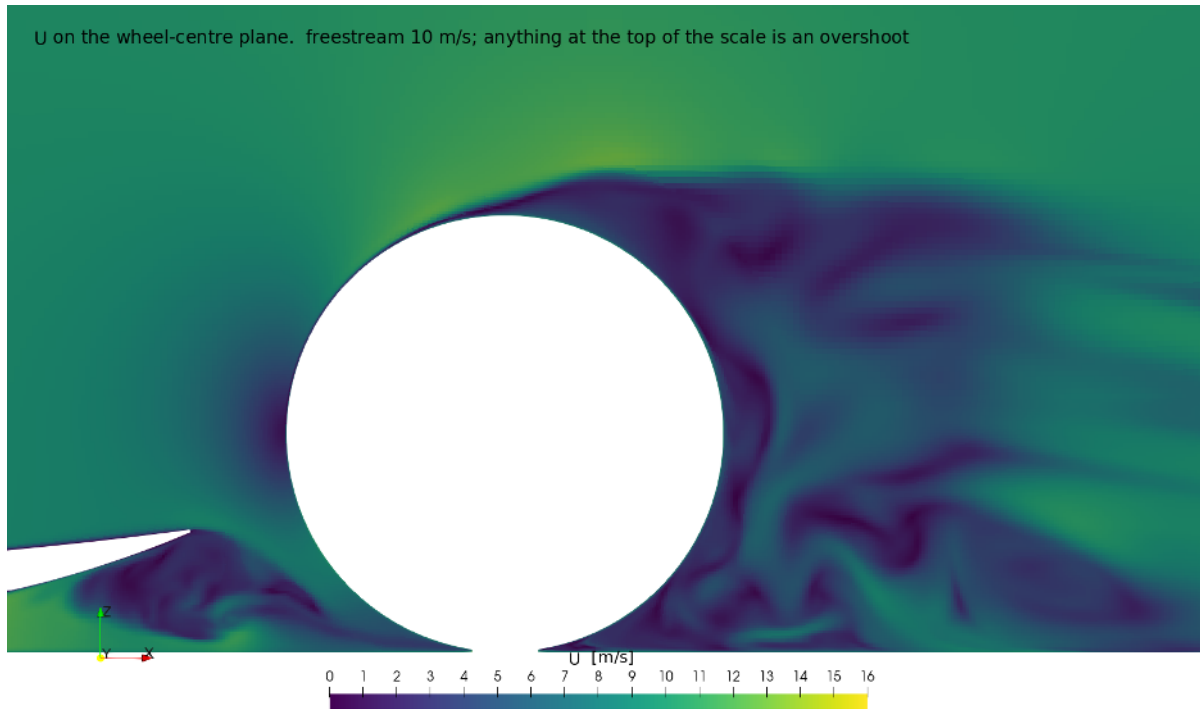
**Figure 16:** Force history. The grey band is the window averaged for the reported result.



**Figure 17:** Settling of the total  $C_L \cdot A$  over the final iterations.

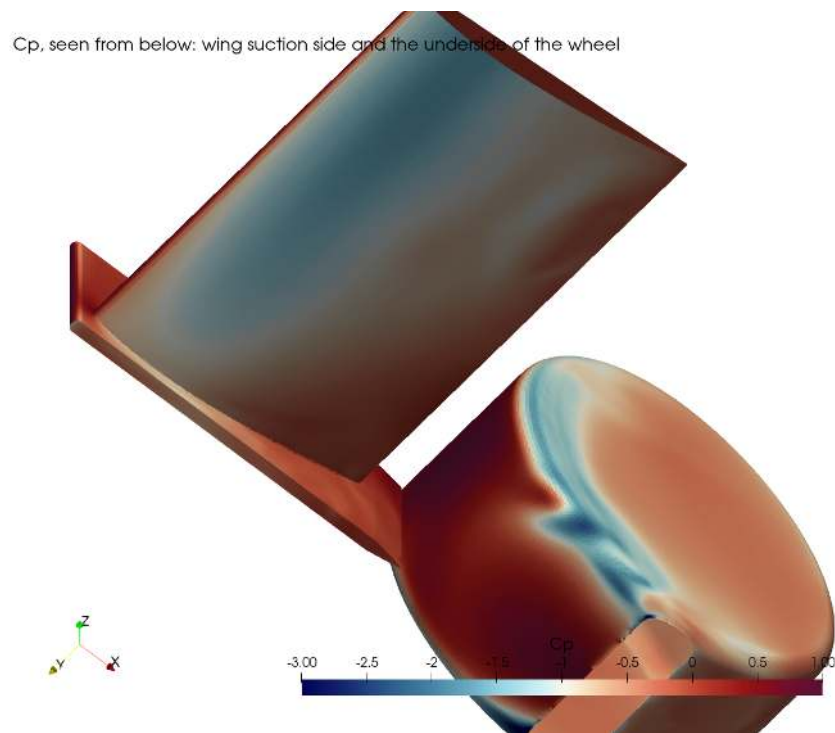
## 8 Results

**Colour ranges are fixed, not auto-scaled.** The raw fields contain isolated extremes far outside the physical range — measured on this build,  $C_p$  down to  $-32$ ,  $C_p T$  up to  $100$  and  $|U|$  up to  $100$  m/s against a  $10$  m/s freestream. Those are single cells, not regions: the velocity figure below shows nothing at the top of its scale. Auto-scaling to them would flatten every plot, so the ranges are the physically meaningful ones and anything outside saturates.

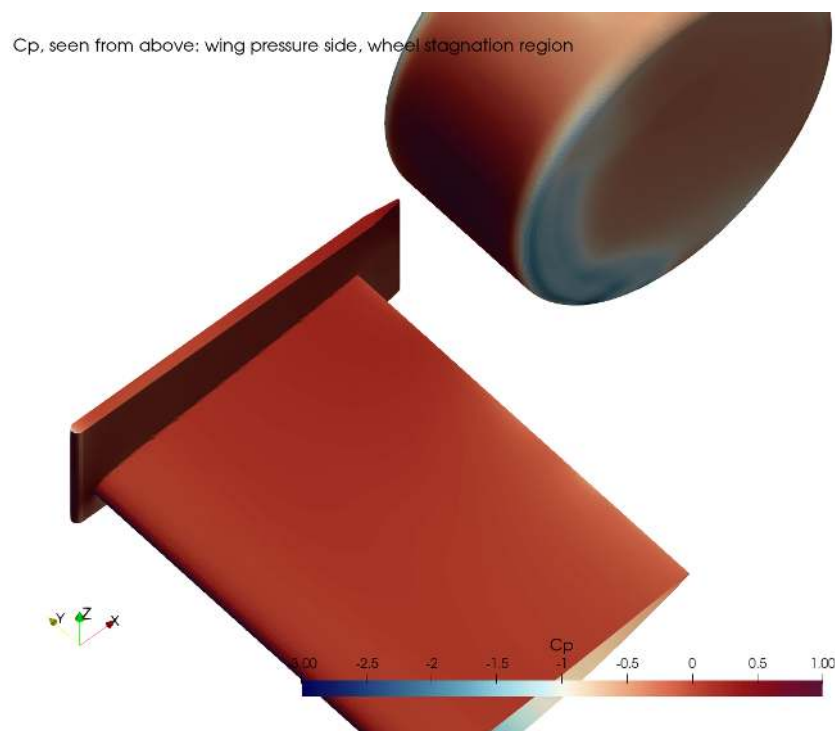


**Figure 18:**  $|U|$  on the wheel-centre plane. Acceleration over the wheel crown, the separated wheel wake, and the wing wake below. Nothing reaches the top of the scale, which is how you tell the extreme values are isolated cells rather than a region.

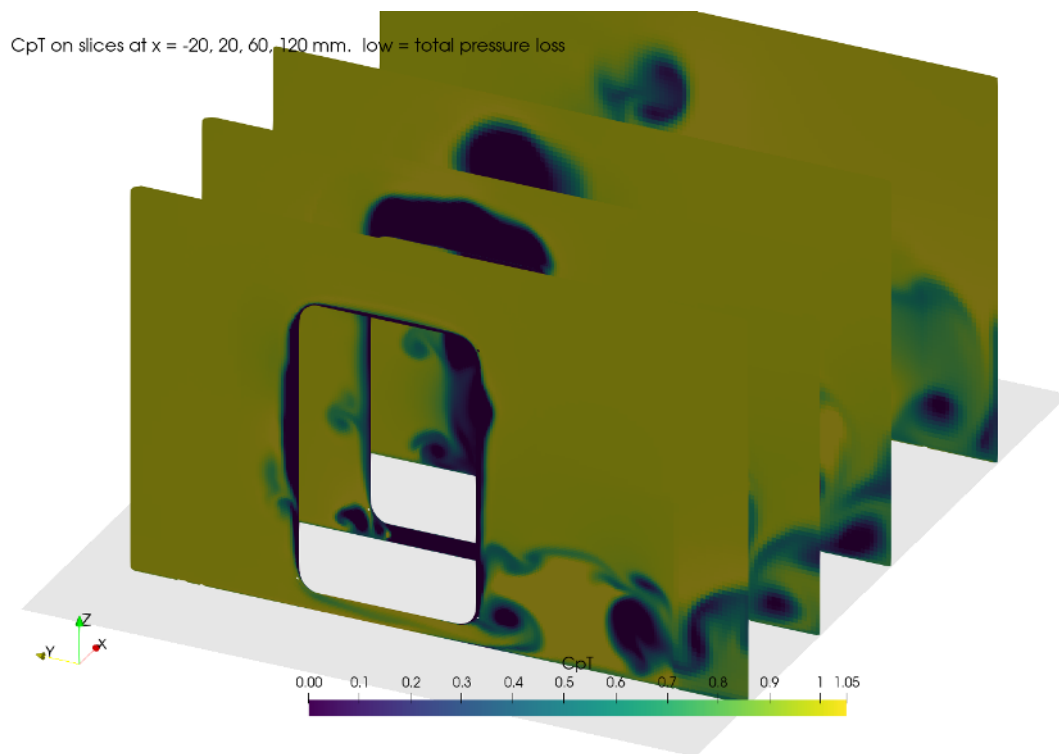
$y^+$ , the check that decides whether the wall treatment holds. Mean values: wing  $3.94$ , wheel  $4.65$ . Per-patch averages on this build sit between  $2.3$  and  $4$  on the wing and between  $2.9$



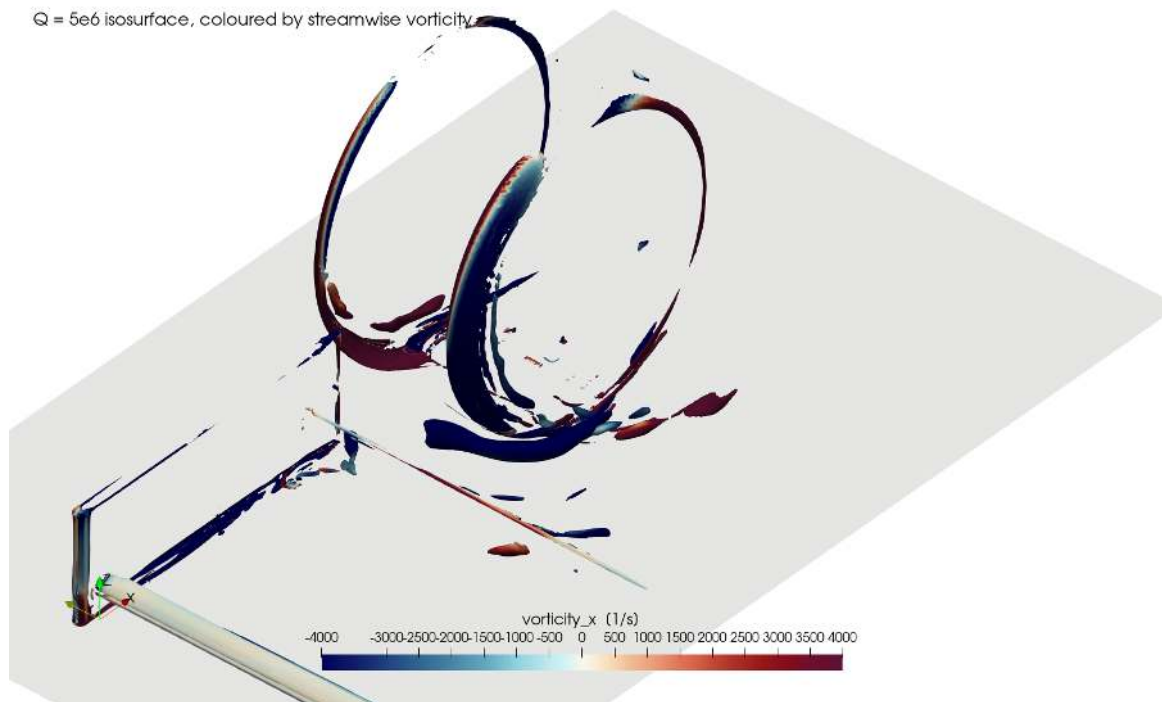
**Figure 19:**  $C_p$  seen from below: wing suction side and the underside of the wheel.



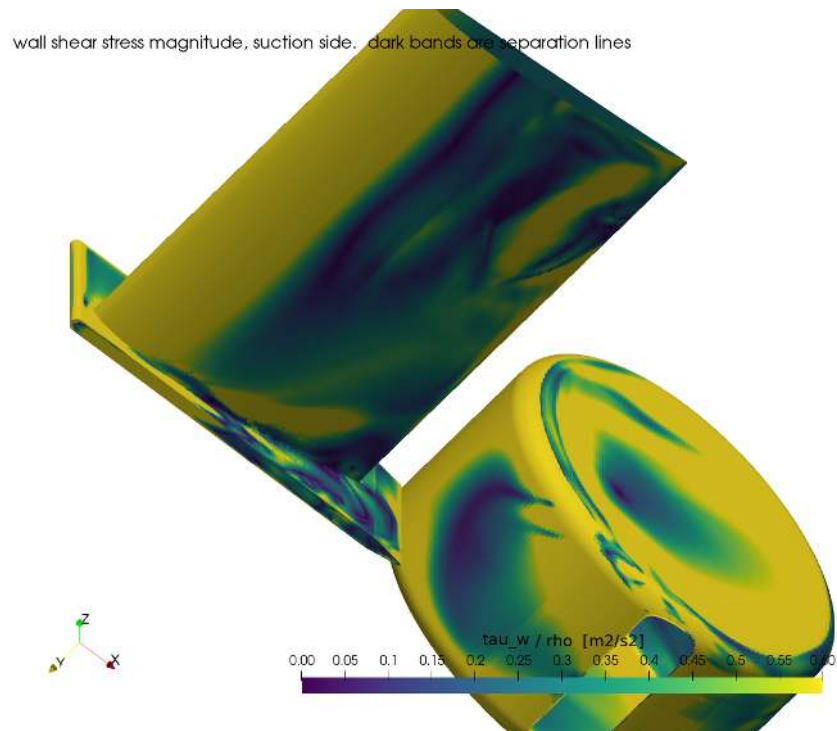
**Figure 20:**  $C_p$  seen from above: wing pressure side and the wheel stagnation region.



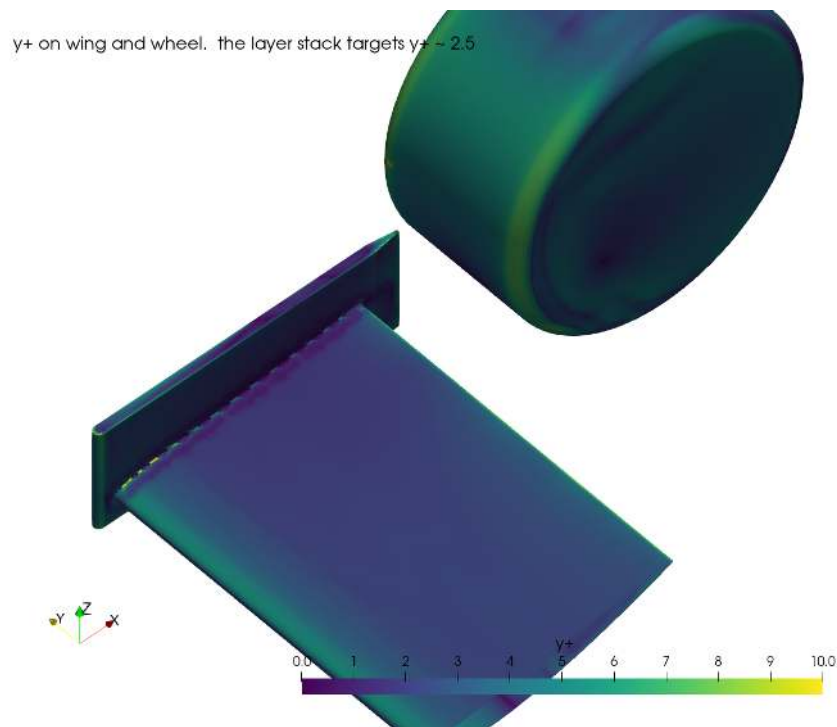
**Figure 21:** Total pressure coefficient on streamwise slices.  $C_p T \approx 1$  in the freestream; the dark regions are total-pressure loss, and the discrete cores downstream are the endplate and wheel vortices.



**Figure 22:** Q-criterion isosurface, coloured by streamwise vorticity — the endplate and wheel vortex systems.



**Figure 23:** Wall shear stress magnitude on the suction side. Dark bands are separation lines.



**Figure 24:**  $y^+$  on the wall patches.

and 3.7 on the wheel tread, shoulders and sidewall — the range the layer stack was sized for. Two outliers, both expected:

- **wheel-plinth** averages 8.7. It carries no prism layers, so the first cell is a bulk cell.
- **ground** peaks at 73.1, in the contact-patch region where the belt passes under the wheel. Local, and not where the forces come from.

## 9 Finishing this report

The solve in this build stopped at 88 of 2000 iterations. To produce the final version:

```
cd RANS_Simulations/S1.42_h0.13_AOA8_W0.63
./Allsolve --continue          # resumes from the latest written time
./Allpost --summary            # refreshes results.json

cd ../../report
./build.sh                     # regenerates every figure, rebuilds the PDF
```

- `-continue` resumes rather than restarting: `startFrom` is `latestTime` in `controlDict`.
- The run stops by itself once `residualControl` is met, which is usually well before the configured 2000.
- To stop it cleanly and keep the state resumable, set `stopAt writeNow` in `system/controlDict` — the solver writes the current state and exits. `runTimeModifiable` is on, so it picks the change up within one iteration. Killing the process instead loses everything since the last write.
- `build.sh -no-figures` rebuilds only the text, which is quick.

Nothing in the report is typed by hand: all numbers come from the case's `results.json` and solver logs via `report/scripts/make_tex_data.py`, so a rebuild picks up the final solution automatically.

## 10 Assessment

**What supports trusting these numbers.**

- **Background mesh exactly as designed** (123 624 cells), cubic, dividing the domain without remainder.
- **Refinement reproduces the previous validated HELYX setup** level for level, and the final cell count agrees with it closely.
- **Patch groups resolve correctly** — `patchSummary` confirms `noSlip` on the wing, `rotatingWallVelocity` on the wheel and the belt velocity on the plinth.
- **Wheel rotation sign verified** analytically against the belt speed.
- **Divergence-free initialisation** via `potentialFoam`.
- **$y^+$  in the design range** of the chosen wall functions.

**What to be careful about.**

- **No curvature correction and no separation fix** relative to HELYX. Expect faster vortex diffusion and possibly different separation onset.
- **Relaxed mesh quality limits** leave genuinely poor cells in the mesh (`minTetQuality` disabled). Accepted in exchange for layer coverage.
- **Three patches without prism layers**, see section 5.
- **Two duplicate triangles** in the wing STL.
- **Steady RANS on a flow that may not be steady**. At high angle of attack or low ride height, check the force spread before quoting a number.

- The two local refinement boxes that stand in for HELYX's per-region distance refinement are **positioned for this geometry** and do not follow the parameters. Re-check them before extending the sweep.